Master 2 – DSC and MLDM - 2024-2025

Data Mining for Big Data - C. Largeron

Documents are not allowed except a handwritten single-sided page
All the solutions have to be justified

Part 1.

We consider a corpus D composed of 6 documents D1, D2, D3, D4, D5 and D6 given below:

D1 — The market impacts the budget, shaping the financial economy
D2 — Economy and finance: opening of the competition
D3 — Today, the sports assembly opens with handball
D4 — Appointment of the director of the budget, finance and economy
D5 — Replacement of the director of the sports assembly by that of handball
D6 — Opening of the session devoted to the market and the economy

In the sequel, we do not make distinction between uppercase and lowercase, we suppose that the stop-words from the list {the, impacts, shaping, and, of, competition, today, with, appointment, replacement, by, that, session, devoted, to} have been removed and we do not consider the punctuation. We also assume that "financial" and "finance" correspond to the same lemma "finance" and "opening" and "opens" to the lemma "open".

Q1. What is the index size for this corpus after stop word removing and lemmatization?

In the sequel, we consider this index and we assume that the words are sorted in alphabetic order in the index.

Q2. What is the inverse document frequency of the index-word 'open' and what is its TF-IDF in document D3?

Q3. What is the term(s) having the lowest IDF? Why?

Q4. We assume that the documents D3 and D5 belong to the category C1 while the other documents of D belong to the category C2. Using the Khi2 statistic and a threshold equals to 1, determine whether the index-word "open" is a discriminant feature of C2.

Q5. Given the new document D7 : " Opening of the assembly with the director of sports and handball", classify this document according to the K nearest neighbor with K = 3, the dot product similarity and the boolean representation of the documents with index of Q1.

Part 2.

We consider another corpus containing 3 documents Doc1 to Doc3 and the document-term matrix (X) associated to these documents using the Boolean scheme and the index I = {T1, T2, T3, T4}.

X	Doc1	Doc2	Doc3
T1	0	1	0
T2	0	0	0
T3	1	0	0
T4	0	1	1

The singular value decomposition of X is defined by the following matrices U and V and the singular values of X are 1.6, 1.0 and 0.6 (values rounded for simplification).

U =

−0.5	0	0.9	0
0	0	0	−1
0	−1	0	0
−0.9	0	−0.5	0

V =

0	-1	0
-0.9	0	0.5
-0.5	0	-0.9

Q6. Propose a graphical representation of the terms and the documents on the two first latent factors.

Q7. Propose a rank one approximation of X and indicate the documents which are correctly represented.